How AI Explanations Influence Trust, Reliance, and Automation Bias

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Updated Project Items:

The primary goal of our research is to discover if explainable answers from generative AI contribute to the cognitive offloading and automation biases already present in users of generative AI. We already know that cognitive offloading and automation bias are persistent trends across users, so understanding contributing factors to these trends is crucial to the responsible and ethical use of generative AI.

Proposed Research Questions:

RQ1: Does the level of explanation accompanying an AI suggestion increase a participant's confidence in the AI helper? **RQ2**: Do participants develop automation bias toward the AI agent over the course of the experiment? Is this affected by the explanation level/accuracy level of the AI helper? **RQ3:** How accurately are participants able to gauge their own accuracy? Does reliance on AI affect their ability to correctly gauge their own accuracy?

Hypotheses:

H1: When using an AI helper with a greater level of explanation, participants will become more confident in the AI-generated suggestions, compared to using an AI helper with a brief or no explanation. **H2**: Over the duration of the experiment participants will develop greater automation bias. **H3**: Participants who are most confident in the AI suggested solutions will lose their ability to gauge their own accuracy.

Proposed Experiment:

Experimental Task: Participants will answer several rounds of SAT-style questions with an AI helper that suggests an answer and an explanation for each question. Before and after each round, participants will rate on a 1–10 scale their confidence in their own answers, their confidence in the AI helper, the difficulty of the questions, and how accurate they think they were. We chose SAT-style questions because each has one clear correct answer and most participants are already familiar with the format.

Apparatus:

A custom program, similar to the one used in Assignment 2, will present the questions and the AI helper's suggestions. The software will also record all survey information regarding user confidence in the suggested answers and their ability to answer the questions. The "AI" will be a scripted set of answers written by our group rather than a live model to ensure every participant sees the same suggestions and explanations.

Independent Variables:

- Explanation level (none, brief, detailed): our main variable of interest, used to test whether more explanation changes how much participants trust the AI.

- AI accuracy: the percentage of correct suggestions in a block of questions. The AI is always correct in the first block and then begins giving some incorrect suggestions, which lets us see whether participants notice the errors. This also serves as a "primer" for automation bias, since automation bias typically only starts to occur after continuous correct answers are given.

Dependent Variables:

- **Score**: shows whether participants follow incorrect AI suggestions.
- **Response time**: shows how much effort participants put into checking answers.
- **Self-reported accuracy**: compared to the actual score to measure how well participants judge their own performance.
- **Confidence ratings**: confidence in themselves and in the AI helper, before and after each round.
- **Experimental Design and Counterbalancing**: We will use a within-subjects design, so every participant will experience all three explanation levels. This lets us compare each participant to themselves, which controls for differences in ability and requires fewer participants than a between-subjects design. All participants will begin with the same control round (no explanation, 100% accurate AI) to record their baseline behavior before the AI starts making mistakes. There are six possible orderings for how users can experience the explanation levels of the model, so each participant will be a unique question ordering. This full counterbalancing keeps learning and fatigue effects from building up in any one condition.

Current Status (20 Points): Please provide an overview and a detailed list of the current status of the items for your proposed experiment. This should include progress on development, the experimental setup, and any other relevant achievements.

So far our team has coordinated a shared github repository along with a kanban task board that aligns with our project. Our team has also collaborated with Claude to build out the README.md and the project architecture inside the repo. Beyond that we have also used agentic development to create the first version of the experiment software. It is a JavaFX desktop application that runs a full participant session from start to finish. The results are currently being saved in a csv file. There are two JSON files, one for controlling the experimental config and the other containing the questions that the participant will be asked. Current status:

1. Completed creating the shared repo.
2. Written the README.md
3. Defined a simple project architecture.
4. Set up the experimental flow (set up -> welcome -> round introduction -> question -> rating -> and thanks).
5. All 6 orderings of the three explanation levels are defined. Each participant is assigned an order from their id number.

Next Goal (10 Points): List the tasks you intend to complete next. How do you plan to finish the remaining incomplete tasks?

1. Build out the question bank. We need to replace the 12 placeholder questions with real questions from the SAT suite. We need them to have clear, correct answers, be solvable in about a minute without a calculator, and have a range of difficulty.
2. We need to decide how we're going to generate AI explanations and generate those explanations.
3. We still need to decide the number of questions per round.
4. We need a google doc for consent, demographics, and short survey about prior AI use, and end of study questions
5. We need to do pilot testing
6. We need to write analysis scripts.
7. Participant recruitment and data collection.

Challenges (8 Points): What challenges are you encountering in completing your project? How do you plan to address these challenges?

One problem we are facing is ensuring that our questions and AI explanations are consistent between participants so we properly test the variables we need. We also need to select an appropriate number of questions that allows us to collect the data necessary without causing participants to have reduced engagement.

Another challenge we have is developing a strategy for prompting the AI helper to ensure that our suggested answers in the experiment are consistent across all participants. We also need to make sure we can clearly distinguish between the levels of explanation for each answer.

We plan to address these challenges by keeping questions and AI responses consistent across participants and conducting pilot test before data collection.

Team-member Responsibilities (10 Points): Outline the responsibilities of every team member.

- Settle the open design decisions (questions per round, accuracy level, time limit, layout, rating wording): **everyone**
- Pick the real questions from the College Board SAT question bank: **Jonah**
- Finalize the question count and accuracy per round:**everyone**
- Write the brief and detailed AI explanations: **Jenny**
- Finalize experiment procedure: **everyone**
    - Experiment Introduction script: **Cameron**
    - Consent and demographics form:**Jonah**
    - Instructions script: **Caleb**
    - Decide if we want to do a short demo: **everyone**

    - Questionnaire: **Cameron**
- Finalize the experiment software and get group approval: **Jenny**
- Look over entire experiment and do a test run to ensure it is complete and behaves as expected:**Jonah**
- Find 2-3 participants and schedule pilot sessions: **everyone**
- Run the pilot sessions and back up the CSVs: **everyone**
- Clean the data and analyze RQ1:**Cameron**
- Clean the data and analyze RQ2:**Caleb**
- Clean the data and analyze RQ3:**Jenny**
- Literature survey:**Caleb**
- Final report, presentation and materials: **everyone**, split later

CITI Training Certificate (0 Point): Upload each team member's CITI training certificate. If any team member lacks the CITI certificate, the team cannot conduct the experiment.

Software Demo (20 Points): Provide a **video demo** of the current stage of the experimental program. The minimum length of the video is 1 minute, and the maximum length is 2 minutes.